



Bar size effect on long-term durability of sand-coated basalt-FRP composite bars

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ABSTRACT

Recently, the use of basalt fiber-reinforced polymer (BFRP) bars as a reinforcing material has gained increasing interest worldwide. In the last decade, the durability of BFRP bars has been the concern of the researchers, however, the effect of bar size or diameter have not yet been addressed. This paper investigates the effect of bar size on the physical, mechanical, and durability characteristics of newly developed sand coated BFRP bars conditioned for 3 months at 60°C in an alkaline solution simulating a concrete environment. Five diameters of BFRP bars (nominal diameters of 9.5, 13 mm, 12.7 mm, 15.9 mm, 19.0 mm, and 25.0 mm) were investigated before and after conditioning. In addition, the microstructure of the BFRP bars—both conditioned and unconditioned specimens—was investigated with scanning electron microscopy (SEM) and Fourier transform infrared spectroscopy (FTIR) to assess any changes or degradation. The results revealed that the physical and mechanical properties of the reference BFRP bars were significantly affected by the cross-sectional size of the bar. The higher probability of defects contained in the larger diameter bars may have caused the lower interlaminar and transverse shear strength, and flexural strength in comparison to the smaller diameter bars. On the other hand, the conditioning in the alkaline solution had a greater negative effect on the mechanical properties of the larger BFRP bars than on the smaller ones. According to the finding of this research, there were variations between the physical and mechanical properties of the BFRB bars of different diameters.

1. Introduction

The use of fiber reinforced polymer (FRP) bars has gained increasing interest worldwide as a structural material for reinforcing new construction because of their corrosion performance in aggressive environmental conditions [1,2]. One of the major applications for FRP bars is to replace conventional steel bars in concrete structures [3,4]. The increasingly high cost of carbon-FRP (CFRP) bars and relatively poor resistance to the alkaline of E-glass-FRP (GFRP) may make the applications of FRPs as structural reinforcement are limited [5]. Advances in fiber-reinforced-polymer (FRP) technology and innovate, however, have spurred interest in introducing new type of fibers—such as basalt fibers—in addition to the commonly used glass, carbon, and aramid. Basalt-FRP (BFRP) is the most recently FRP composite materials developed to enhance the safety and reliability of structural systems compared to GFRP, CFRP, and AFRP composites [6]. BFRP characterizes superior mechanical properties and is more stable chemical resistance

compared with E-glass FRP. BFRP also demonstrates a wider range of working temperatures and provide significantly better cost-effectiveness compared to carbon-FRP (CFRP) [7]. The basalt fibres are an environmentally friendly and non-hazardous material as they are directly produced from volcanic rocks originating from frozen lava, with a melting temperature between 1500°C and 1700°C [8,9].

The durability of glass, carbon, and aramid FRP reinforcement in civil engineering environment, especially in normal concrete environment has been investigated by many researchers [1,10–20]. As a result, material specifications and the current design guidelines [21–24] have also been developed to encourage the construction industry to use GFRP, CFRP, and AFRP as the main reinforcement in concrete structures and provide design recommendations for using these bars. Only limited work, however, was studied on durability and relevant applications of BFRP bars because the technology is relatively recent compared to other FRP composites [2,7,8,25]. Besides, a limited FRP design codes or guidelines have incorporated the BFRP as an FRP alternative. The

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mechanical, physical, and durability characteristics of newly BFRP reinforcement are, therefore, important information needed in the material specifications and design codes or guidelines of FRP bars. These properties should be determined in accordance with the prescribed test standards and methods [21,22]. They must also be made available to design engineers, asset owners, and building-code and standard-writing authorities [26].

Basalt-FRP composite bars are made from continuous basalt fibers impregnated in resin using the pultrusion process. The previous studies showed that the common type of resins are used in manufacturing BFRP bars such as vinyl-ester and epoxy [2,27]. They are non-corrosive, consist of 80% fibers and have a tensile strength three times that of the steel bar normally used in building construction. Currently there are many BFRP bar manufacturing companies which market their products, such as Galen LLC and Magma-Tech Ltd Company. Like steel reinforcement, BFRP bars available in diameters ranging from 6 mm to 25 mm, larger diameter bars can be also manufactured. Unlike steel reinforcement, whose properties can be assumed to be the same for different bar diameters, FRP bars are size dependent in terms of longitudinal strength due to shear-lag effect [28]. A reduction in tensile strength of up to 40% was measured by Holloway [29] when the bar diameter increased from 9.5 mm to 20 mm. Therefore, the BFRP manufacturers are required to fully report the characteristic strength, stiffness, physical, and durability properties for every type and size of BFRP bar, especially in harsh environmental conditions. As internal reinforcement, FRP bars are embedded in a cementitious environment aggressive to basalt FRPs (BFRPs) due to the high pH level of the pore-water solutions and the presence of alkali ions [15,26,30,31].

There have been few studies investigating the effect of alkaline environment on physical and durability properties of BFRP bars with different bar size. Benmokrane et al. [30] investigated the physical, mechanical, and durability characteristics of three different types of FRP bars made of basalt and glass fibers with vinylester and epoxy resins. The test results revealed that the Glass/Vinylester composite had the best physical and mechanical properties and lowest degradation rate after conditioning in alkaline solution. A total of 132 BFRP specimens comprising two types and seven diameters were tested by Serbescu et al. [32] in tension after conditioning in pH9 and pH13 solutions at 20, 40, and 60°C for 100, 200, 1000, and 5000 h. The tested BFRP bars exhibited a tensile strength ranged from 972 to 1481 MPa while the elastic modulus varied from 34 to 47 GPa. The lowest strength retention of 69% was observed for BFRP bars exposed to pH9 solution at 60°C for 5000 h. Li et al. [25] investigated the effect several parameters, such as temperature, bonded length, diameter and type of FRP bars, concrete strength, concrete cover thickness and a fireproof coating material on the concrete surface, on the bond strength of the BFRP bars. The results revealed that the bond strength declined gradually with the increase of the temperature, and the effect of temperature on the bond strength of the GFRP bar pullout specimens was more severe than that of the BFRP bar pullout specimens. At temperatures from 70°C to 220 °C, the bond strength decreased by 2.45–14.24%. At 270°C, the bond strength declined by approximately 32%. At 350°C, the residual bond strength was 12.20–23.14% of their original values. Most recently, Ali et al. [2] investigated the physical, mechanical, microstructural, and durability characteristics of newly developed BFRP bars. The durability performance of the BFRP bars was then assessed by conducting the mechanical tests, such as transverse-shear test, flexural test, and interlaminar-shear test, on the specimens after different exposure periods (1000; 3000; and 5000 h) at 60°C. The test results revealed that the unconditioned BFRP bars had the best physical properties. On the other hand, the long-term durability performance revealed that the transverse shear-strength, flexural-strength, and interlaminar shear-strength retention were decreased by 12%, 19%, and 21%, respectively.

This paper systematically investigates the physical and mechanical characterization of BFRP bars of different diameters ranging from 9.5 mm to 25 mm. It aims to correlate the physical and long-term

characteristics of FRP bars with bar diameters for their effective use and to provide guidance in the development of design codes and material specifications for this reinforcing material. In the first and second stage of the study, the physical and mechanical characterization of the unconditioned bar materials were determined. In the third stage, the BFRP bars were exposed to alkaline solution for 3 months at 60°C to simulate the concrete pore environment. Physical and Mechanical characterization of these conditioned bars was then conducted and compared to that of the unconditioned bars. Microstructural analyses and measurements of the physicochemical properties were also carried out on the conditioned and unconditioned BFRP bars.

2. Experimental program

This experimental work was conducted to investigate the short- and long-term characteristics of newly developed BFRP bars. This research work is a part of an extensive research project being conducted at the University of Sherbrooke through NSERC Research Chair activities to develop and introduce new FRP reinforcement as internal reinforcement for concrete structures especially that subjected to harsh environmental conditions, such as marine structures, parking garages, and bridge deck slabs. The findings of this research project will contribute to integrating BFRP into FRP standards and guides, such as ACI 440 [21], CSA S807 [22], and CSA S806 [23], CSA S6 [24], ACI 440 [33].

The research program was divided into three phases. Phase I focused on physical characterization of the BFRP bars. The physical properties determined in this phase served as references for physical properties after conditioning. Phase II focused on mechanical characterization of the BFRP bars. The tensile strength, tensile modulus of elasticity, transverse-shear strength, flexural strength, flexural modulus of elasticity, interlaminar-shear strength, and bond strength were determined according to the appropriate test methods. The test results also served as references for calculating the residual strengths after conditioning. Phase III assessed the durability and long-term performance of the conditioned BFRP bars. The durability was assessed by immersing the BFRP specimens in an alkaline solution at high temperature (60°C) for 3 months (90 days) designed to simulate a concrete environment so as to validate the performance of the BFRP bars as internal reinforcement for concrete elements. Changes in the physical and mechanical characteristics were assessed by comparing the characteristics of the conditioned BFRP bars to the reference ones from Phases I and II.

2.1. Material and test specimens

The sand-coated ROCKBAR™ basalt-FRP (BFRP) bars used in this study were made of continuous basalt fibers impregnated in a vinyl-ester resin and were manufactured according to the pultrusion process by a Russian company (Galen LLC., Russia). Five diameters of basalt-FRP bars were investigated (#3, #4, #5, #6, and #8), which correspond to nominal diameters of 9.5 mm, 12.7 mm, 15.9 mm, 19.0 mm, and 25.0 mm, respectively. The nominal cross-sectional area of the 5 bars is 71 mm², 127 mm², 198 mm², 285 mm², and 491 mm² for bar size #3, #4, #5, #6, and #8, respectively, as reported by the manufacturer. The mechanical properties reported in the current study were calculated using the nominal cross-sectional area. Fig. 1a shows the BFRP bars. For each bar size, the preparation of specimens and characterization of the physical and mechanical properties of the BFRP bars were performed according to the appropriate ASTM, ACI, and CSA test standards. Table 1 summarizes of the specimens dimension, number of specimens tested, and test standard methods.

The tests, in physical characterization, were conducted in accordance with ACI 440 [21] and CSA S807 [22] and the relevant ASTM standards. The cross-sectional area of the BFRP bars was assessed according to CSA 806 [23], Annex A, basalt-fiber content according to ASTM [34], transverse coefficient of thermal expansion according to ASTM [35], moisture absorption according to ASTM [36]. The cure ratio

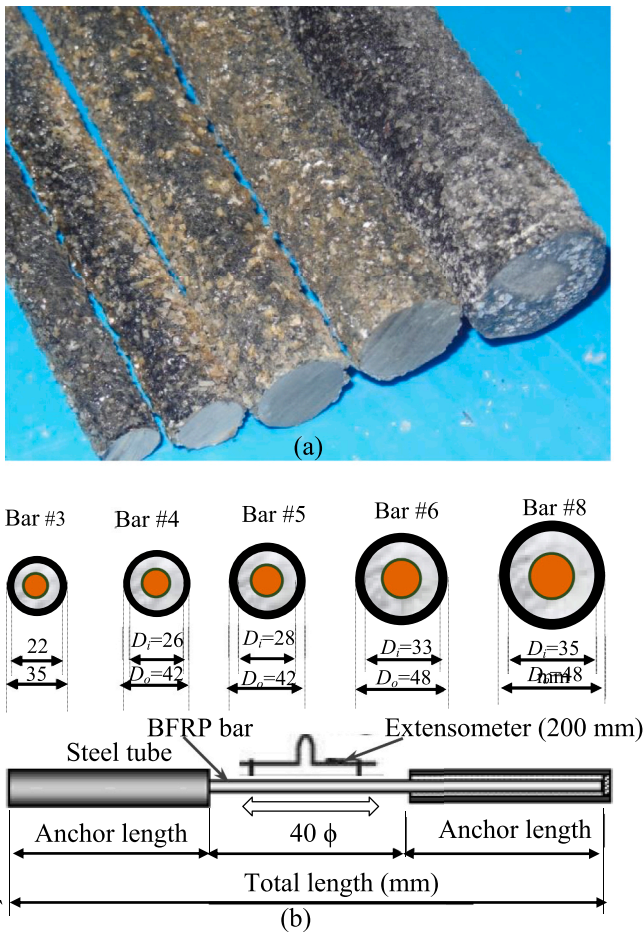


Fig. 1. (a) Basalt FRP bars; (b) specimen preparation for tensile test.

and glass transition temperature, T_g , of the BFRP bars were determined by differential scanning calorimetry (DSC) according to the ASTM [37] test method. In addition, microstructural analysis was performed for five diameters of BFRP specimens using scanning electron microscopy (SEM) for both the unconditioned (reference) and conditioned specimens to assess changes and/or degradation. The effects of conditioning on the glass-transition temperatures (T_g) and on BFRP-bar chemical composition were also determined with differential scanning calorimetry (DSC) and Fourier transform infrared spectroscopy (FTIR), respectively. Table 2 summarizes the results of the physical characterization of bars.

The mechanical characterization of BFRP bars with different diameters were included testing of representative bars to determine their tensile strength in accordance with ASTM [38] and CSA-S806 [23] Annex C, transverse shear strength in accordance with ASTM [39]; interlaminar shear (short-beam test) in accordance with ASTM [40];

flexural strength in accordance with ASTM [41], and bond strength in accordance with ACI 440 [21], B.3 and CSA-S806 [23], Annex H. The nominal diameter and nominal cross-sectional area of the BFRP bars were used in calculating the mechanical properties. Table 3 summarizes the results of the mechanical characterization of bars.

2.2. Environmental conditioning

To simulate the concrete environment, the alkaline solution was prepared according to the recommended formula by ACI 440 [21] and CSA S806 [23]. The formula requires 1 L of deionized water mixed with 118.5 g of $\text{Ca}(\text{OH})_2$, 0.9 g of NaOH , and +4.2 g of KOH . The pH of the alkali solution was 12.8, which is representative of a mature concrete pore solution. It is worth noting that the alkaline environment in concrete has a pH above 12 [33]. The BFRP specimens were completely immersed in alkaline solution at 60°C and were removed from the solution after 3 months (90 days). The conditioned BFRP bar samples were removed from the solution and tested under tension to compare their tensile-strength retention values to those of the reference specimens. The conditioning temperature was set at 60°C, as specified in ASTM [38]. During conditioning, the level of alkaline solution and pH level were checked periodically, and new solution added as necessary.

2.3. Mechanical characterization of the BFRP bars

2.3.1. Tensile properties

The tensile properties of the investigated BFRP bars were assessed according to ASTM [38] and CSA S806 [23], Annex C. The tensile testing along the alignment of fibers is related to fiber properties. The gauge length of the specimens was approximately equal to 40 times the bar diameter, in addition to the length of the anchorage steel tubes at each end of the BFRP bars, as specified in CSA S806 [23], Annex B (see Fig. 1b). The steel tubes (anchors) were attached to BFRP specimens before conditioning. Table 1 provides the specimen and anchor length for each bar size. The specimens were instrumented with a 200 mm LVDT to capture specimen elongation during testing, as shown in Fig. 2. To avoid damaging the LVDTs, they were detached from the specimen when the load reached 60% of the estimated ultimate load. The tests were conducted with a Baldwin testing machine in which the applied load and specimen elongation were electronically recorded during the test with a computerized data-acquisition system. Fig. 2 shows the test setup. The load was increased at a rate of 300 MPa/min until tensile failure occurred. Four specimens were tested for each bar diameter. This test determined the ultimate tensile strength f_t , tensile modulus E , and tensile strain ϵ .

2.3.2. Transverse-shear strength

The transverse shear is the major structural force on dowels in jointed pavements or on stirrups in concrete beams. Transverse-shear tests were conducted according to ASTM [39] to characterize the tested bars. The setup consisted of a 230 × 100 × 110 mm steel base

Table 1
Summary of the specimens dimension; number of tested specimens, and test standard methods for each bar size.

	Bar size	Type of mechanical tests for reference and conditioned specimens				
		Tensile test		Flexure test	Interlaminar shear test	Transverse shear test
		Gauge length	Anchor length			
Length of specimen (mm)	#3	460	500	175	60	160
	#4	580	550	180	70	170
	#5	700	600	200	85	180
	#6	850	700	210	100	195
	#8	1000	850	240	125	210
No. of specimens	For each size	8 ^a		8 ^a	8 ^a	8 ^a
Standard test method	For each size	ASTM [38]		ASTM [41]	ASTM [40]	ASTM [39]

^a 8 BFRP specimens (4 specimens as references; 4 specimens as conditioned at 60 °C after 3 months).

Table 2
Results for physical properties of BFRP bars and the specified limits for FRPs.

Property	Bar designation					Specified limits	
	#3	#4	#5	#6	#8	ACI 440 [21]	CSA S807 [22]
Nominal diameter (mm)	9.5	12.7	15.9	19.0	25.0	–	–
Nominal cross-sectional area (mm ²)	71	127	198	285	491	–	–
Actual cross-sectional area by immersion test (mm ²)	91	129	257	398	550	–	–
Fiber content (%)	80.4	80.3	82.6	82.4	79.6	55 (vol %)	70 (wt%)
Transverse CTE, (x10 ⁻⁶ /°C)	18.1	19.0	15.0	18.6	21.7	NA	40
Water absorption at 24 h (%)	0.21	0.50	0.41	0.17	0.19	–	0.35 (D2 ^b); 0.25 (D1 ^a)
Water absorption at saturation (%)	0.44	0.63	0.70	0.21	0.41	1.0	1.0 (D2 ^b); 0.75 (D1 ^a)
Cure ratio (%)	100	100	100	100	100	NA	≥93 (D2 ^b); 95 (D1 ^a)
T _g (°C)	133	130	126	122	118	100	≥80 (D2 ^b); 100 (D1 ^a)
Wicking dots	0	0	0	0	0	–	–

^a Classification based on durability: FRPs with high durability shall be classified as D1; FRPs with moderate durability shall be classified as D2; FRP made with vinyl ester and epoxy shall be classified as D1 or D2; FRPs made with polyester matrix shall be classified as D2.

equipped with lower blades spaced at 50 mm face to face, allowing for the double transverse-shear failure of the specimen caused by an upper blade, as shown in Fig. 3a. For each size of bar tested, five unconditioned specimens were tested under laboratory conditions on an MTS 810 (MTS Systems Corporation, Eden Prairie, Minneapolis) testing machine equipped with a 500 kN load cell. A displacement control rate of 1.3 mm/min was employed for the #3, #4, and #5 bars, while 1.8 mm/min was used for the #6 and #8 bars until specimen's failure. The loading was done without subjecting the test specimens to any shock. The transverse-shear strength was calculated as shown in Eq. (1).

$$\tau_u = P_s / 2A \quad (1)$$

Table 3
Mechanical properties of the reference BFRP bars.

Bar designation	Mechanical properties					
	Tensile strength (MPa)		Transverse shear strength (MPa)	Interlaminar shear (MPa)	Flexural properties	
	f_u (MPa)	E_u (GPa)			Flexural strength f_{uf} (MPa)	Modulus of elasticity E_f (GPa)
#3	1240 ± 66.9	58.4	231 ± 16.1	58 ± 0.9	1367 ± 6.7	63.8
#4	1050 ± 18.0	48.0	214 ± 11.9	52 ± 6.3	1093 ± 88	54.2
#5	1388 ± 11.9	56.2	217 ± 6.1	53 ± 2.7	1421 ± 10	64.1
#6	1508 ± 16.5	63.0	194 ± 10.5	42 ± 2.5	1527 ± 16	69.5
#8	1023 ± 31.8	52.9	188 ± 5.6	49 ± 2.8	1174 ± 28	60.5

where τ_u = transverse-shear strength (MPa); P_s = failure load (N); and A = bar cross-sectional area (mm²).

2.3.3. Interlaminar-shear strength

In pultruded FRP bars in which the fibers are arranged unidirectionally and bonded with the polymer matrix, the horizontal stresses would be more conducive to inducing interface degradation than transverse-shear stresses [42]. The short-beam shear testing was conducted following ASTM [40] in order to determine the interlaminar shear strength (ILSS) of FRP bars (see Fig. 3b). The test was carried out with MTS 810 testing machine equipped with a 500 kN load cell. Five replicates for each bar diameter were prepared and tested. The distance between the shear planes was set to 6 times the nominal diameter of the BFRP bars. A displacement control rate of 1.3 mm/min was employed for the #3, #4, and #5 bars, while 1.8 mm/min was used for the #6 and #8 bars. The applied load and displacement was recorded with a computer-monitored data-acquisition system. The interlaminar shear strength S_u of the BFRP bar was calculated as shown in Eq. (2).

$$S_u = 0.849P / d^2 \quad (2)$$

where P is shear failure load (N); and d is bar diameter (mm).

2.3.4. Flexural properties

For quality control and specification purposes, the flexural testing is especially useful. Flexural properties may vary with specimen diameter, temperature, weather conditions, and differences in straining rates. The test was conducted in accordance with ASTM [41]. The unconditioned specimens were tested over a simply supported span equal to 20 times the bar diameter and with an overhang twice the bar diameter at each support. Five replicates for each bar diameter were tested under laboratory conditions on an MTS 810 testing machine equipped with a 500 kN load cell. The specimens were loaded at the mid-span with a circular nose and supported at the ends on two circular supports that allowed the specimens to bend, as shown in Fig. 4. A displacement control rate of 3.0 mm/min was used. The applied load and deflection were recorded during the test with a data acquisition system monitored by a computer. The flexural strength of tested FRP specimens was calculated as follows:

$$f_u = PLC / (4I) \quad (3)$$

where P is the failure load, L is the clear span, c is the distance to the centroid of the extreme-most fibers, and I is the moment of inertia.

2.3.5. Bond strength

The bond strength of the BFRP bars was assessed with the pullout test. The pullout tests were carried out in normal-strength concrete with a designated strength of 35 MPa after 28 days according to the ACI 440 [21], B.3 Test Method and CSA S806 [23], Annex G. The bonded length was kept constant at $5d_b$, where d_b is the nominal diameter of FRP bars.

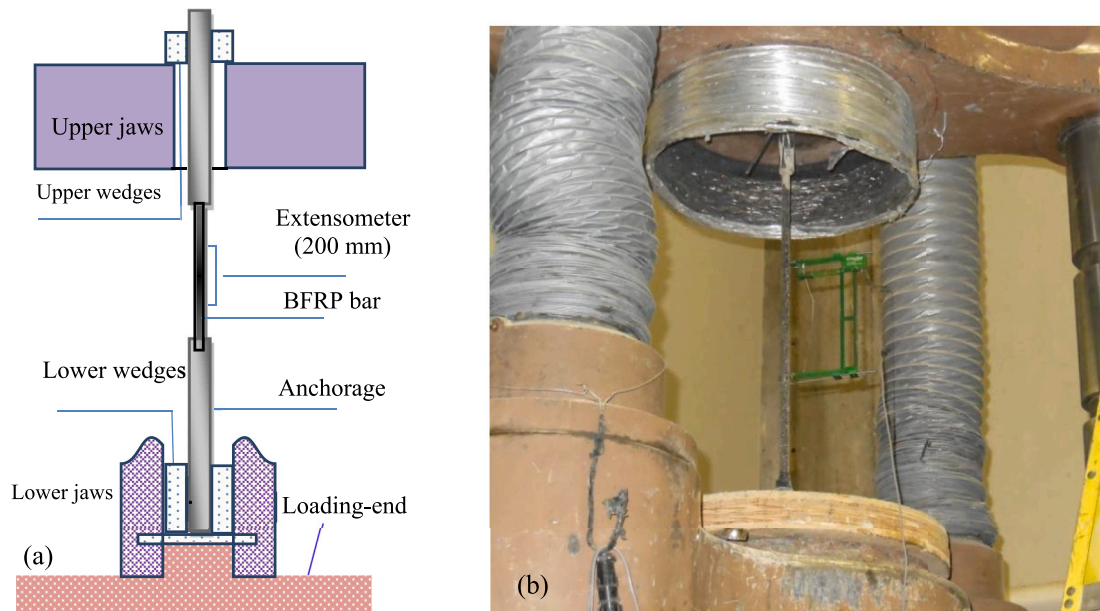


Fig. 2. Tensile test set-up: (a) Schematic configuration (b) Specimens in the machine.

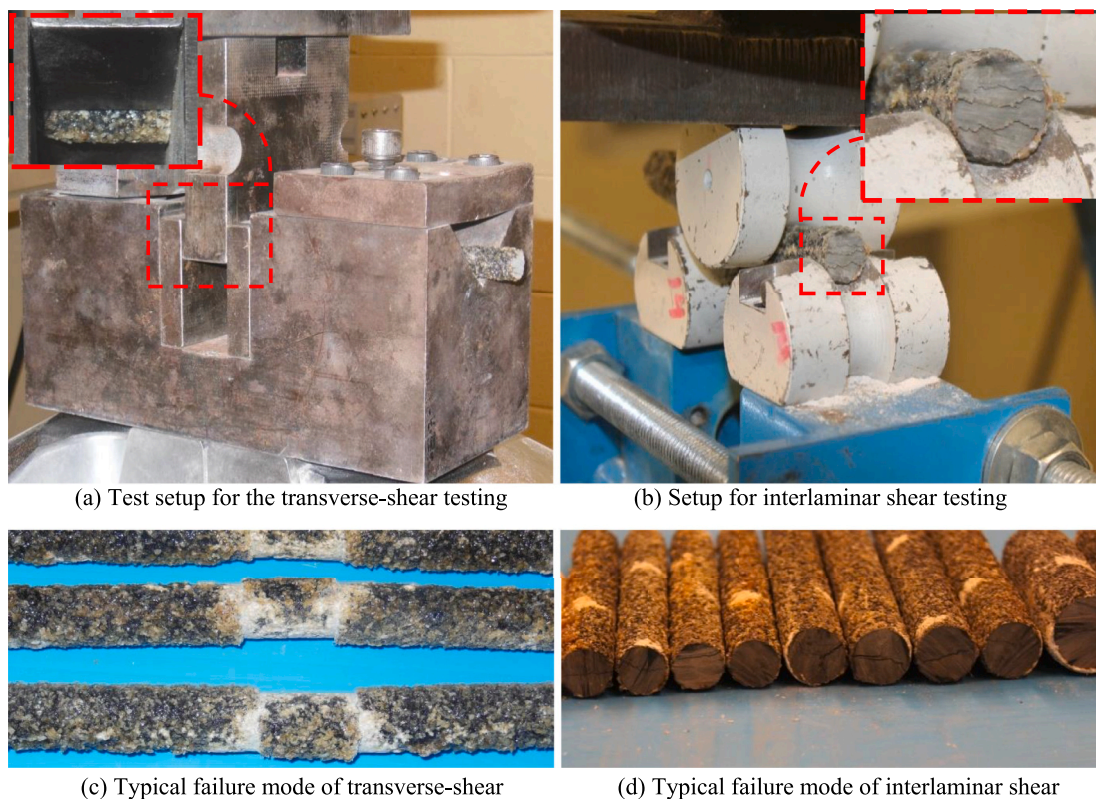


Fig. 3. Test setup for transverse and interlaminar shear testing.

The pullout blocks measured $200 \times 200 \times 200$ mm (for bars #3, #4, and #5) and $300 \times 300 \times 300$ mm (for bars #6 and #8). Fig. 5 shows the geometry of the pullout specimens, test setup, and modes of failure, while Fig. 8 shows the typical bond stress-free slip relationship of the tested BFRP bars.

3. Results and discussion

The analysis and discussion for the effect of bar size on the physical,

mechanical, and durability characterization of BFRP bars are presented in this section.

3.1. Effect of bar size on physical characterization

For each bar size, the physical characterization was performed according to ACI 440 [21] and CSA S807 [22] test methods. Since BFRP bars are not included in any FRP standard yet, the physical characterization of the investigated BFRP bars was compared to the specified

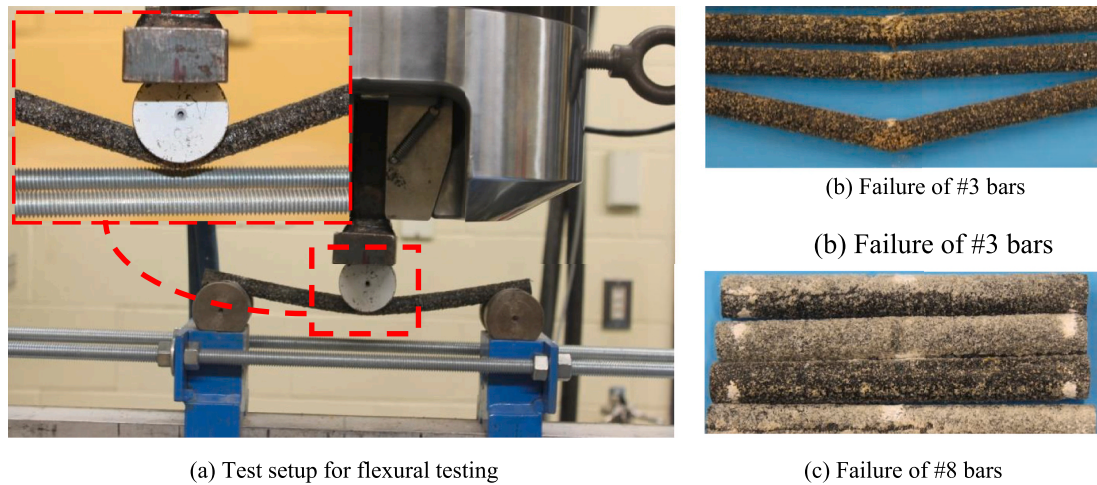


Fig. 4. Typical flexural-strength test and specimens at failure.

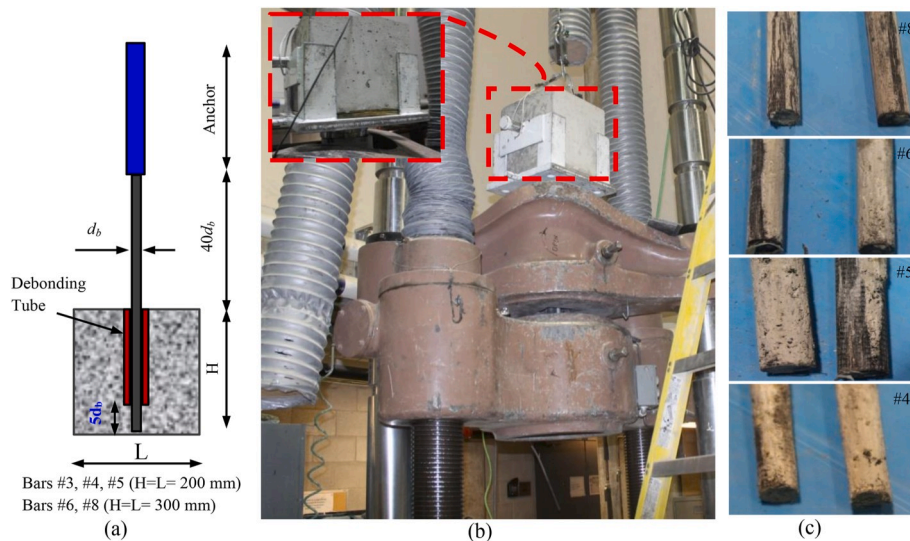


Fig. 5. Pullout test. (a) Specimen geometry; (b) Test setup; (c) Typical pullout bond failure.

limits for FRP bars. Table 2 presents the results of physical characterization of tested BFRP bars and the limitations specified in ACI 440 [21] and CSA S807 [22] for FRP bars. The results in Table 2 showed that the bar size had no significant effect on some of the physical characterization of the BFRP bars, including the transverse coefficient of thermal expansion (Transverse CTE), and T_g . Similarly, a cure ratio of 100% was measured for all bar size, indicating that bar size did not affect the degree of cure. This is attributed to the development of an efficient production method of BFRP bars. This is in good agreement with the results of Benmokrane et al. [43], however, in contrast with the observations made by Yi and Hilton [44]. Who showed that laminate thickness might affect the degree of cure due to the higher thermal conductivity of thicker composite laminates. The maximum value of transverse CTE observed with the BFRP bar size (#8), due to the lower fiber content (higher resin content), which is less than the limits of $40.0 \times 10^{-6}/^{\circ}\text{C}$ as stated by CSA S807 [22].

On the other, the water absorption and fiber content were observed to increase and decrease with increasing bar size, respectively, as shown on Table 2. The lower water absorption was in the large diameter of bars, since none of the BFRP bars contained voids. This can be correlated to increasing fiber content in large diameter of bars. Basalt fibers absorb scarcely the water; therefore, the BFRP bars with higher matrix contents

evidenced higher absorption rates. Moisture diffuses into all organic polymers, leading to changes in thermophysical, mechanical, and chemical characteristics. The primary effect of the absorption is on the resin itself through hydrolysis, plasticization, saponification, and other mechanisms, which cause both reversible and irreversible changes in the polymer structure. In some cases, the moisture wicks along the fiber matrix interfaces and has been shown to cause deleterious effects to the fiber-matrix bond, resulting in a loss of integrity at that level. In Table 2, the results also showed that the water absorption of the BFRP bars with different bar size at saturation was satisfied the limits specified in CSA-S807 [22] and ACI 440 [21] (1%).

In order to further correlate bar size to the percentage water absorption, the shape ratio of the BFRP bars were determined and plotted, as shown in Fig. 6. The shape ratio is defined by Cinquin and Medda et al. [45] as the ratio between the sample's surface and volume as reported in Table 3. As can be seen in Fig. 6, the shape ratio was significant in the water absorption of the BFRP bars at 24 h (24 h) and at saturation, i.e., the water absorption increased as did the shape ratio. It can also be observed that the relationship between the water absorption at 24 h (%) and at saturation to that of the shape ratio is the same, as demonstrated by the almost equal slopes of the water absorption and shape-ratio relationship curves. On the other hand, there is a linear but negative

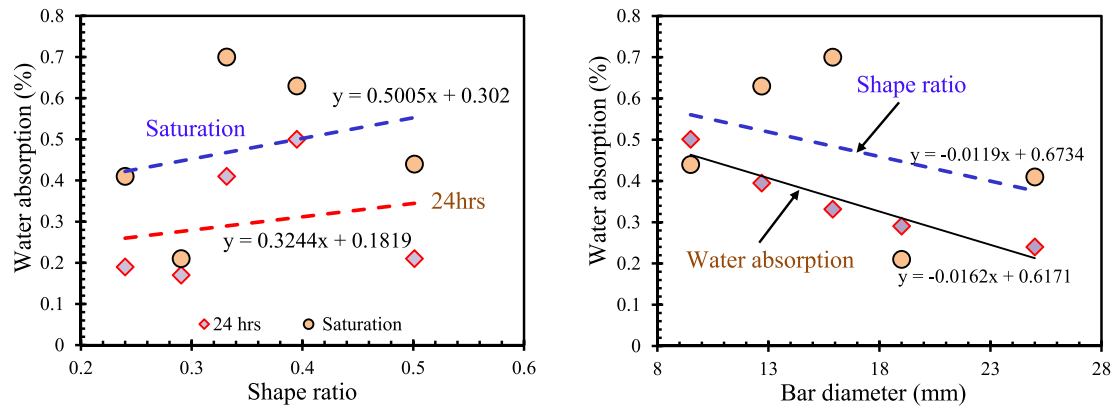


Fig. 6. Relationship of water absorption to the shape ratio and bar diameter.

correlation between the shape ratio and bar diameter. The shape ratio decreases as the bar diameter increases. This accounts for the smaller diameter bars (#4) having higher absorption rates than the larger diameter ones (#8) with the same length, since the exposed surface is greater with respect to volume. It is also worth noting that the decrease in the shape ratio is very similar to the decrease in the percentage of water absorption at saturation (%) for the various bar diameters. It should be mentioned that no wicking was detected in all BFRP bars. The fiber content of the BFRP bars was 80.4%, 80.3%, 82.6%, 82.8%, and 79.6% which correspond to the bar sizes of 9.5 mm (#3), 12.7 mm (#4), 15.9 mm (#5), 19.0 mm (#6), and 25.0 mm (#8), respectively, thereby satisfying the ACI 440 [21] (55% by volume) and CSA S807 [22] limits (70% by weight). The BFRP bars were oversized when compared to their respective nominal cross-sectional areas. The average measured oversize varied from 2% to 39%, as seen with the BFRP bar (#4) and (#6), respectively. This variation is due to the ratio difference of the sand coating to the core.

3.2. Effect of bar size on mechanical characterization

Generally, tensile properties of FRP bars are among the most important parameters affecting the design of FRP-reinforced concrete sections, since tensile strength governs section capacity and the tensile modulus of elasticity governs the serviceability limit state. Numerous of studies, on the durability of FRP, have revealed that the short-term mechanical characterization of FRP bars decrease with increased bar diameter [5,28]. This conclusion was clearly found in our study. Fig. 7 presents the normalized mechanical characterization with the different sizes of BFRP bars. The figure provides the percentage of the tensile strength, transvers strength, interlaminar-shear strength (ILSS), flexural strength, and tensile modulus for all of the bar sizes with respect to the mechanical characterization of the #3 bars. It can be seen that, significant difference in the tensile strength of the BFRP bars regardless of bar size. Highest tensile strength and modulus were observed for the #6 bars, the lowest tensile strength were exhibited by the #4 bars (84% compared to the #3 bars), with the #8 bars exhibiting more than 86% of strength and stiffness of the #3 bars. This result is in contrast with the observations made by Kocaoz et al. [46] and Benmokrane et al. [43], who suggested that modulus, which is an intrinsic property of the material, is not significantly affected by bar cross-sectional size but rather by the level of fibers contained in the bar. Since the fiber content's percentage for the different bar size was almost the same, the elastic modulus of the bars should be the same value. Furthermore, the consistency in the resulted tensile characterization of the BFRP bars with different bars size can be due to an adequate anchor length, which resulted in a more efficient transfer of stresses from the bar surface to the center. To avoid this phenomenon, Portnov and Bakis [47] suggested that introducing a more uniform distribution of the applied shear stress

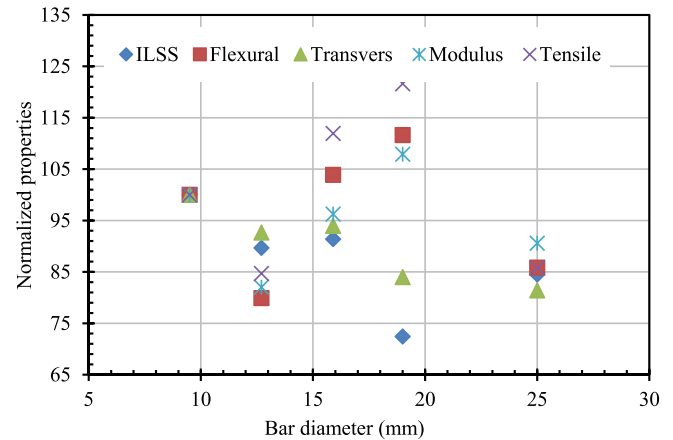


Fig. 7. Normalized mechanical properties of the BFRP bars.

near the grips could minimize the shear lag effect and increase the tensile strength of FRP bars.

In similar behavior of tensile properties, Fig. 7 indicated that there was a size effect observed for transvers strength, ILSS and flexural strength. The BFRP #8 bars showed transvers strength, ILSS and flexural strength almost 18%, 15%, and 14% lower than the BFRP #3 bars. Table 4 also showed that the highest transverse-shear strength was observed for the #3 bars, the lowest transverse strength were exhibited by the #8 bars. It is worth mentioning that these results were expected, since the transverse-shear strength resulted mainly from the resin with a small contribution from the fiber-resin interface [20]. For ILSS's results, the #8 bars recorded almost 15% lower than the #3 bars, This was observed by Wisnom and Jones [48], significant size effects were observed on the ILSS for unidirectional GFRP composite bars. They concluded that the lower ILSS with bigger specimens could be due to the larger defects inherent defects. Koller et al. [49] also suggested that the probability of having large defects in FRP composite bars increases with increasing bar size. Moreover, it should be noted that ILSS is a matrix-dominated property. Wisnom and Jones [48] indicated that matrix-dominated failures show the largest size effects in composite

Table 4
Sample geometry and shape ratio.

Bar size	Diameter, <i>d</i> (mm)	Sample length, <i>L</i> (mm)	Shape ratio, SR
#3	9.5	25.0	0.50
#4	12.7	25.0	0.39
#5	15.9	25.0	0.33
#6	19.0	25.0	0.29
#8	25.0	25.0	0.24

materials. Defects near the edge of the GFRP bars were very critical for the specimens subjected to the short-beam test as this location is subjected to higher levels of shear stress. Similarly, the lower flexural strength of the #8 bars, can be explained by the high probability of defects. Carvelli et al. [50] suggested that it is more difficult to keep the filaments parallel to one another in larger diameter FRP bars during the pultrusion process, increasing their tendency to buckle under compression. This is, in fact, the failure mechanism observed in the BFRP bars during the flexural test, in which the failure was initiated by the compression buckling of the top fibers.

Table 4 presents also the bond strength of the tested BFRP bars. It can be seen that, the bond strength of bars obtained herein is ranging from 12 ± 0.7 MPa to 16 ± 1.4 MPa, which is higher than 8.0 MPa as specified by CSA S807 [22]. Thus, it could be concluded that the tested bars meet the mechanical-property requirements of CSA S807 [22]. All of the BFRP specimens failed in a typical pullout mode of failure by exhibiting slip through the free end with no signs of splitting cracks appearing on the cubes (Fig. 8).

3.3. Strength retention of BFRP bars with different bar size

The durability performance of the BFRP bars with different bar size was assessed by conducting the mechanical tests, such as tensile test, transverse-shear test, flexural test, and interlaminar-shear test, on the specimens conditioned in an alkaline solution simulating concrete pore solution. Accelerated aging tests were conducted in accordance with ASTM [38]. The conditioning of the BFRP bars included combined exposure to an alkaline environment and elevated temperature. All of BFRP bars were immersed in the alkaline solution at 60°C for 3 months (90 days). The tested BFRP bars were weighed and their diameters measured throughout the conditioning period to monitor water absorption and eventually characterize the mass and diameter changes. Observation revealed no changes in diameter during the conditioning periods. All specimens were removed from the solution and tested to determine their physical and mechanical properties after the exposure periods at 60°C. The effects of conditioning on the glass-transition temperatures (T_g) and chemical composition were also assessed with differential scanning calorimetry (DSC) and Fourier transform infrared spectroscopy (FTIR), respectively. In addition, the microstructure of all bar size was investigated with scanning electron microscopy (SEM) for both conditioned and unconditioned cases to assess changes and/or degradation.

Fig. 9 and Table 5 provide the strength and stiffness retention of the BFRP bars with different bar size. Figs. 9a, c, 9d, and 9e clearly indicate that the strength properties were affected by conditioning in the alkaline

solution, while Fig. 9b shows that the changes in the tensile modulus of elasticity were not significant for the BFRP bars (less than 2.0%). The almost 100% retention of the tensile modulus of elasticity, which is a fiber-dominated property, for all bar size indicates that the damage caused by the moisture diffusion was confined to the very thin layer of the exposed surface and that the reinforcing fibers were not affected by the conditioning. As the tensile modulus of elasticity was calculated from the linear-elastic portion of the stress-strain curve of FRP bars, the matrix was still effectively transferring stresses to the fibers. As shown in Fig. 9a, the BFRP of #4 bars were highly affected by accelerated aging with an ultimate tensile strength retention of 67%. The higher degradation in tensile strength of the #4 bars is probably related to the high increase in moisture absorption over time, leading to plasticizing effects of the polymer matrix [20]. Nkurunziza et al. [51] indicated that the diffusion of moisture and alkalis in composites can destroy the bond between the fiber and matrix, damaging the interface. This is difficult to see at lower loads, but higher mechanical loads increase the degradation of the fiber-matrix interface. This accounts for the noticeable decrease in strength properties, as these values were calculated at the point of failure of the BFRP bars at which the stress distribution along the fibers is less uniform. On the other hand, the BFRP #3, #5, #6, #8 bars were less affected by accelerated aging with ultimate tensile strength retentions of 82%, 85%, 84%, and 80%, respectively. The results indicated that the #3 bars retained their tensile strength more than those of the #8 bars. This trend can be due to the shear lag effect. Achillides and Pilakoutas [52] indicated that larger diameter FRP bars had more significant shear lag-effect than smaller diameter bars and had a noticeable effect on tensile strength. Castro and Carino [53] further mentioned that the efficiency of the stress transfer from the bar surface to the interior fibers influences the mechanical properties of the FRP bars. Due to the shear lag effect, the outer fibers experience higher stresses than the inner fibers, even though all fibers resist the tensile load at the same time. Due to conditioning in alkaline solution, the outer surface of the bars was affected and might have decreased mechanical properties. As the specimens were loaded, the outer fibers—which were subjected to higher stress—initiated failure and the breakage moved instantly to the inner fibers, leading to the sudden and catastrophic failure of the BFRP bars.

Fig. 9c shows the transverse-shear strength of the BFRP bars after conditioning. The BFRP #4 bars were highly affected by accelerated aging and had a transverse-shear strength retention of 81.7% after 3 months (90 days). The #6 bars were slightly affected by accelerated aging and evidenced strength retentions of 95%, followed by #5, and #3, which had 5.5% and 9.6% degradation in the transverse-shear strength due to conditioning, respectively. Fig. 9d shows that the effect of conditioning on the ILSS of the BFRP bars was significant with the smaller diameter bars, but decreased as the diameter increased. Clearly, the BFRP of #8 bars had the highest interlaminar-shear strength (91%) and the lowest rate of degradation (9%) due to conditioning, followed by the BFRP of #6, #5, and #3. This indicates that the #8 bars had a better fiber-matrix interface which minimized the degradation due to conditioning. While the BFRP of #4 bars retained 77.5% of their ILSS. This can be explained by the effect of diffusion of moisture and alkaline solution in composites which destroy the bond between the fiber and matrix, leading to damaging the interface. As indicated in the previous researches, ILSS is governed by the fiber-matrix interface. Thus, the decrease in ILSS can be correlated to interface degradation.

Fig. 9e shows that the flexural-strength retention increased with bar diameter. The retention ranged from 80% retention for the #3 bars to more than 91.3% for the #8 bars. This behavior was also observed by Maranan et al. [54], who found that larger diameter bars exhibited a slower rate of strength degradation at high temperature than small diameter bars. Cinquin and Medda [45] also concluded that the residual mechanical properties were more affected in thin composites than thick composites. The higher flexural strength retention for the larger diameter bars could be due to the strength gradient through bar diameter during bending. During flexural tests, the outermost fibers are subject to

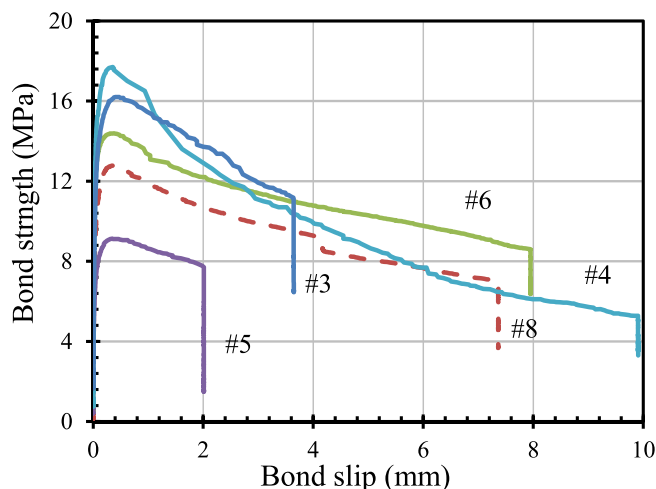


Fig. 8. Bond stress-free slip relationships of the tested BFRP bars.

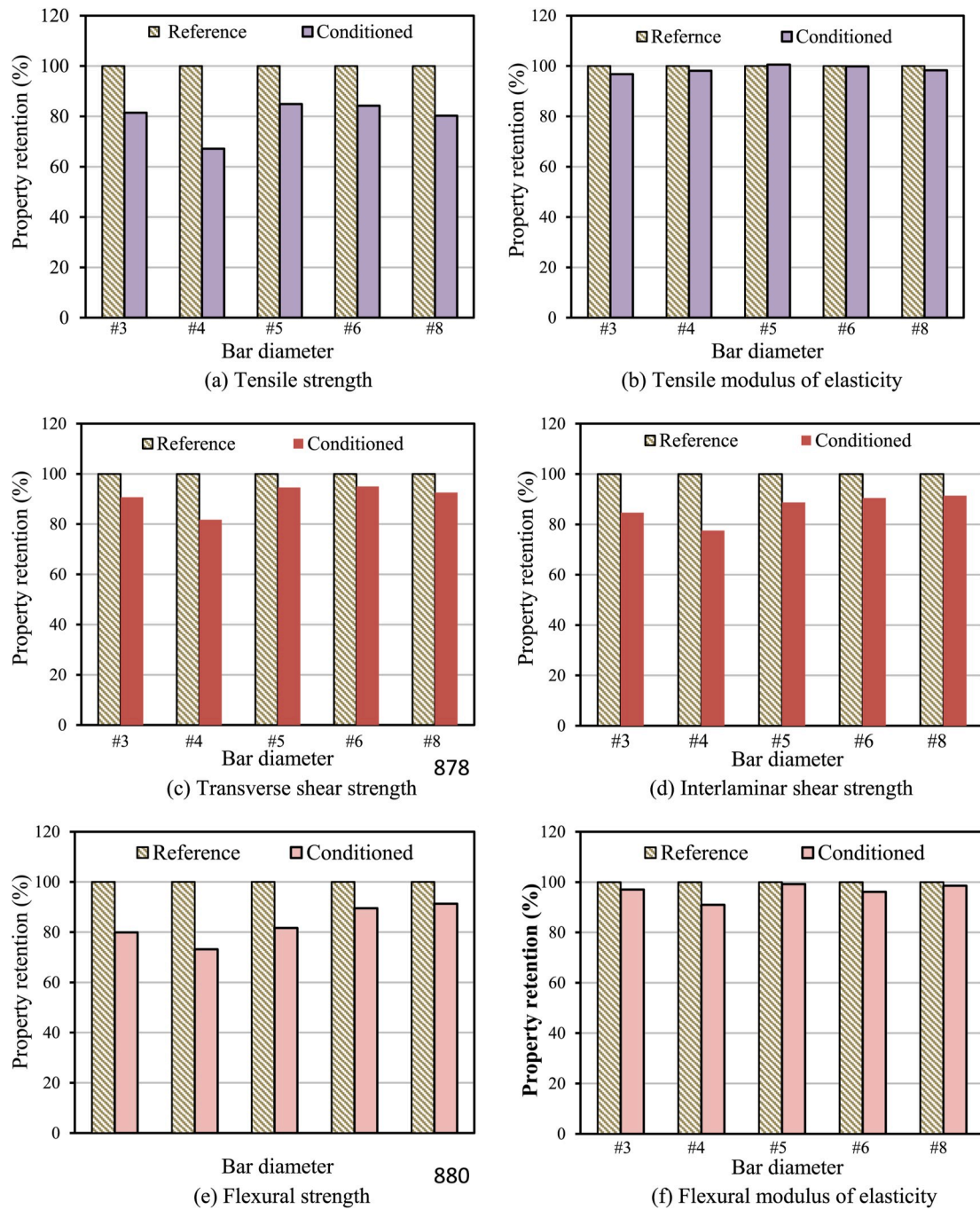


Fig. 9. Property strength retention of BFRP bars with different diameters.

Table 5

Mechanical properties of the conditioned BFRP bars.

Bar designation	Mechanical properties					
	Tensile strength (MPa)		Transverse shear strength (MPa)	Interlaminar shear (MPa)	Flexural properties	
	f_u (MPa)	E_u (GPa)			Flexural strength f_{uf} (MPa)	Modulus of elasticity E_f (GPa)
#3	1010 ± 56.0	56.5	209 ± 11.1	49 ± 5.1	1093 ± 14	62.0
#4	705 ± 2.4	47.0	175 ± 8.4	40 ± 3.6	800 ± 7.4	49.3
#5	1178 ± 17.1	56.5	205 ± 0.7	46 ± 3.3	1160 ± 28	63.6
#6	1270 ± 33.8	62.9	184 ± 4.4	38 ± 2.5	1367 ± 11	66.8
#8	821 ± 16.3	52.0	174 ± 6.0	45 ± 1.3	1072 ± 29	59.6

the maximum stress. Thus, a smaller amount of fibers with respect to the total volume was under the maximum stress in the larger diameter bars. Moreover, the higher flexural-strength retention in the larger diameter bars can be correlated to the smaller reduction in the second moment of area of the conditioned specimens. Since only the bar surface was damaged, the fibers and matrix in the intact zone were undamaged and had the same initial mechanical properties, resulting in a better redistribution of load when the outer fibers progressively failed.

3.4. Moisture absorption and DSC measurements of the conditioned BFRP bars

The moisture uptake at saturation was determined for the specimens conditioned in the alkaline solution at 60°C for 3 months. The moisture-uptake ratios at saturation for the reference specimens were 0.24%, 0.63%, 0.57%, 0.21%, and 0.62% for the BFRP #3, #4, #5, #6, and #8 bars, respectively. The moisture-uptake ratios at saturation for the conditioned specimens were 0.26%, 0.69%, 0.59%, 0.2%, and 0.64% for the BFRP #3, #4, #5, #6, and #8 bars, respectively. The glass-transition temperature (T_g) of the conditioned BFRP bars was determined in accordance with ASTM [37]. One scan was performed for each bar diameter. The conditioned BFRP specimens showed T_g values of 139, 136, 115, 153, and 111°C for BFRP of #3, #4, #5, #6, and #8 bars, respectively. The DSC measurements performed on the conditioned BFRP bars show that the materials are fully cured.

3.5. SEM and FTIR observations

Scanning-electron-microscopy (SEM) observations were performed to investigate microstructural changes in the BFRP bars before and after conditioning in 60°C alkaline solution for the longest duration, which was 3 months (90 days). All of the specimens observed under SEM were cut, polished, and coated with a thin layer of gold–palladium using a vapor-deposit process. The microstructural observations were carried out on a JEOL JSM-840A microscope. Similarly, Fourier Transformed Infrared Spectroscopy (FTIR) was conducted for each bar diameter to study the changes in the chemical composition of the matrix at the bar surface. These observations were conducted to determine the potential degradation of the polymer matrix, glass fibers, or interface, as applicable, due to the penetration of the alkaline solution. The aim was to link these observations to the possible evolution of mechanical properties and chemical composition of the bars after conditioning.

3.5.1. SEM observations

Fig. 10 shows the SEM micrographs of the cross section of the reference and conditioned BFRP of #3, #4, #5, #6, and #8 bars. As shown in Fig. 10, there were pores observed in the center of the reference BFRP bars. The degree of porosity was relatively high with the reference BFRP of #4 bars, compared to those BFRP bars (#3, #5, #6, and #8). All of reference BFRP bars, there was also noticeable gap at the fiber–matrix interface, indicating excellent adhesion between the fibers and matrix, except, the BFRP of #4 bars which presented a slight debonding at the interface between the fibers and matrix.

SEM analysis of the conditioned bar specimens indicates that small gaps between the basalt-fiber and matrix were observed near the exposed surface in the conditioned BFRP of #4 specimens, suggesting that the bars were affected after exposure to the alkaline solution. This damage to the fiber–matrix interface reduces the transfer of loads between fibers and results in the reduction in strength properties. These results can account for the highest moisture uptake measured at saturation for the BFRP of #4 bars. Benmokrane et al. [19], however, reported that the fiber component controls the mechanical properties of FRP bars. If the fibers have not deteriorated, FRP bars will preserve most of their mechanical strength and will be able to support loads. If the protective resin degrades, the bonding between the matrix and fibers located on the outer part of the bar will gradually reduce and bar

resistance will start to decrease. This is the most probable reason why the strength properties of BFRP bars decreased after alkaline conditioning for 3 months. The micrographs of conditioned BFRP of #3, #5, and #6 specimens revealed no significant issues with the fiber–matrix interface as shown in Fig. 10. On the other hand, the conditioned BFRP of #8 specimens presented a slight debonding at the interface between the fibers and vinyl-ester resin (Fig. 10).

3.5.2. Effects on the matrix

Fourier transform infrared spectroscopy (FTIR) was conducted for each bar diameter to identify any chemical change/degradation of matrix at the bar surface after 3 months of conditioning at 60°C. These observations were conducted to determine the potential degradation of the polymer matrix, basalt fibers, or interface, as applicable, due to the penetration of the alkaline solution. The aim was to link these observations to the possible evolution of mechanical properties and chemical composition of the bars after conditioning. Fig. 11 shows the infrared spectroscopy (FTIR) spectra of the surface of the reference and conditioned BFRP bars. Only the FTIR for the #3 and #8 bars are provided for comparison. The observation of FTIR for both bar diameters shows that no clear differences between the infrared spectra of the reference and conditioned BFRP bars were observed as shown in Fig. 11. Moreover, the FTIR did not show any significant changes in the chemical structure when the specimens were immersed in alkaline solution. These observations indicate that the surface of the bars exposed to the alkaline solution for 3 months at 60°C had not been chemically modified and might possess the same initial mechanical properties. These results further confirm that the degradation of the matrix remains only at the exposed surface of the BFRP bars. Nkurunziza et al. [51] confirmed that the vinyl ester epoxies are very resistant to chemical attack, which improves the environmental resistance of FRP bars made with these resins.

4. Summary and conclusions

The effects of diameter on the physical, mechanical, and durability properties of BFRP bars were investigated. Based on the results of this study, the following conclusions were drawn for the tested BFRP bars:

- 1 All of BFRP bars were oversized when compared to their respective nominal cross-sectional areas. The average measured oversize varied from 2% to 39%, as seen with the BFRP bar (#4) and (#6), respectively. This variation is due to the ratio difference of the sand coating to the core.
- 2 With the bar sizes considered, the test observation indicates that, bar diameter was affected fiber content, transverse coefficient of thermal expansion, or glass transition temperature. The water absorption of the BFRP bars with different diameters, at 24 h, is well within the allowable specified limit in CSA-S807 [22], except, the BFRP bars of #4, #5, and #8 which obtained exceed the limits specified in CSA-S807. Improvement is needed.
- 3 The tensile strength modulus of elasticity of the reference BFRP bars were significantly affected by the cross-sectional size. The size effect was also observed for interlaminar shear strength, transverse shear strength, and flexural strength of the reference BFRP bars. The higher probability of defects contained in the larger diameter bars may have caused the lower interlaminar shear strength, transverse shear strength, and flexural strength in comparison to the smaller diameter bars.
- 4 The bond strength of BFRP bars, with different diameters, obtained herein is ranging from 12 ± 0.7 MPa to 16 ± 1.4 MPa, which is higher than 8.0 MPa as specified by CSA S807 [22].
- 5 The interlaminar shear strength, transverse shear strength, and flexural strength of the larger diameter BFRP bars were less affected after exposure to the alkaline solution than the smaller bar diameter. The higher strength retention for the larger bar sizes was due to the lower affected thickness. As a result, the

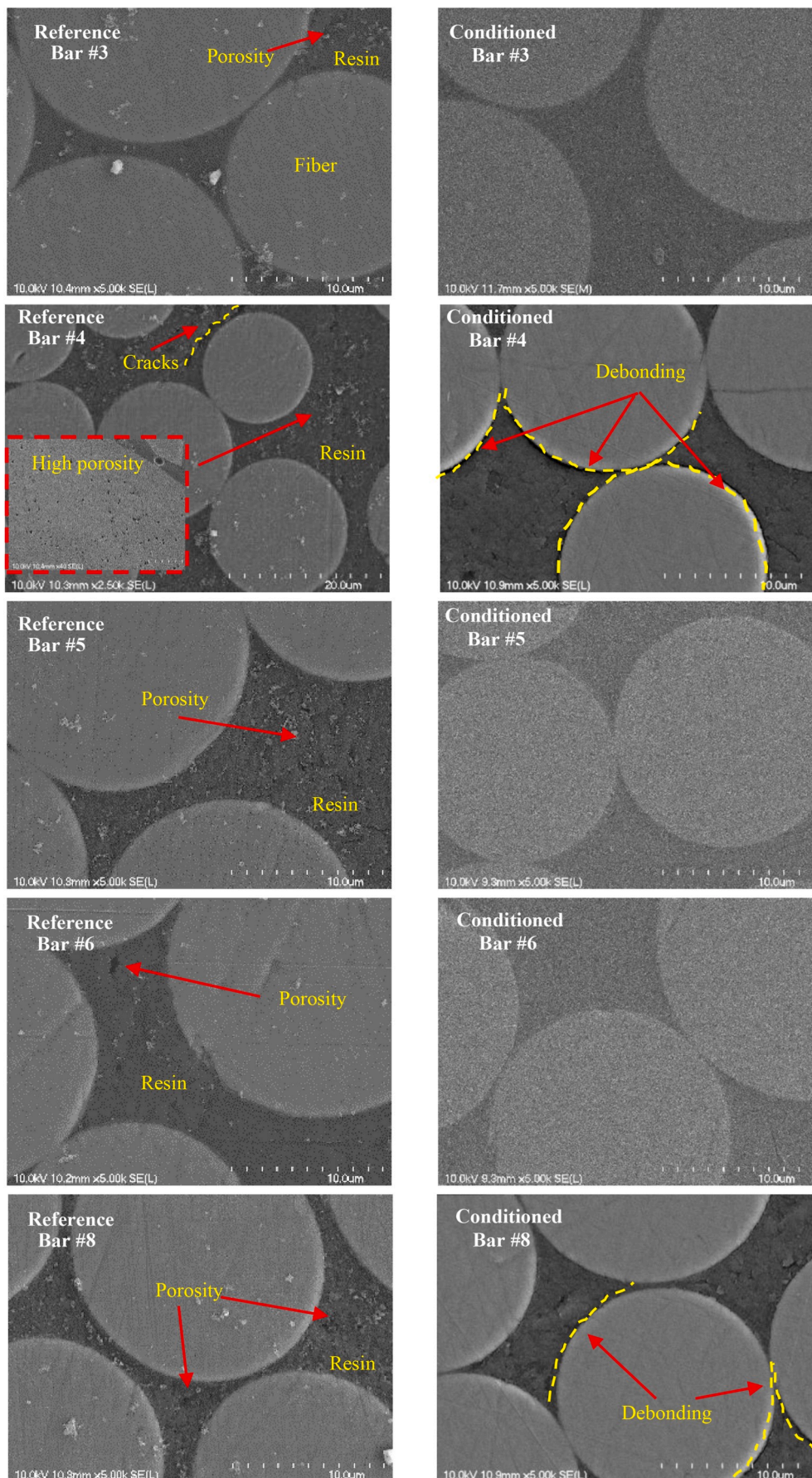


Fig. 10. Micrograph of the unconditioned and conditioned BFRP of the different diameters.

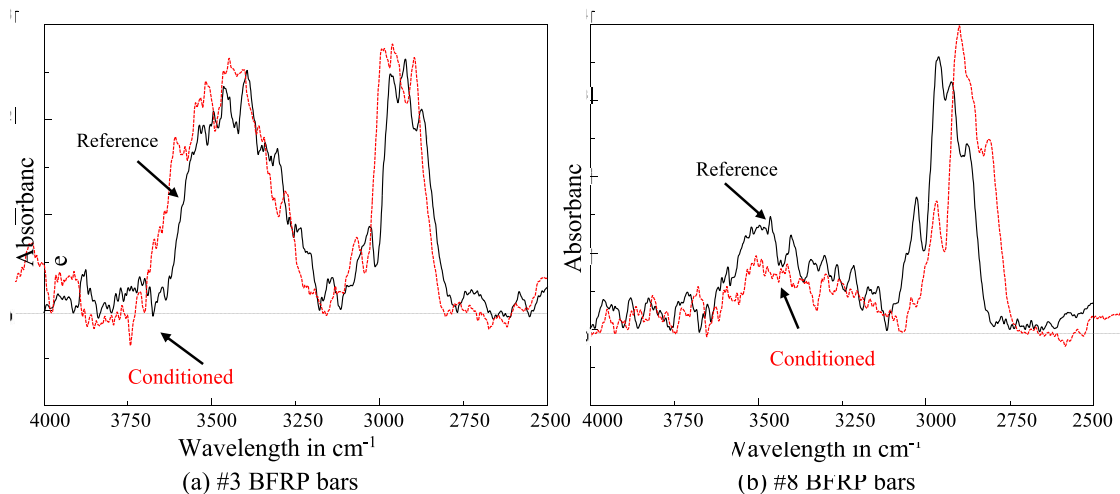


Fig. 11. FTIR spectrum of the BFRP bars before and after conditioning.

penetrated area was proportionally small relative to the total cross-sectional area of the bar.

- 6 The tensile-strength retention was highest for the smallest diameter bar. This suggests that the impact of conditioning on the tensile properties of BFRP bars is expected to be greater for larger than smaller diameters.
- 7 The scanning-electron-microscope (SEM) showed that the pores were observed in the center of the reference BFRP bars. The degree of porosity was relatively high with the BFRP of #4 bars, compared to other diameters. All of reference BFRP bars, there was also noticeable gap at the fiber–matrix interface, indicating excellent adhesion between the fibers and matrix, except, the BFRP of #4 bars which presented a slight debonding at the interface between the fibers and matrix.
- 8 Small gaps between the fiber and matrix were observed near the exposed surface in the conditioned BFRP of #4 specimens, suggesting that the bars were affected after exposure to the alkaline solution.
- 9 The FTIR analysis showed no changes in the material properties and chemical structure in the exposed surface of the bars after conditioning in the alkaline solution for 90 days at 60°C.
- 10 This study has been conducted on sand sand-coated basalt-FRP to study the effect of bar size and diameter. More experimental work and investigations need to be conducted to evaluate the effect of bar surface (helically wound spiral outer surface, indented, braided, or with ribs), rein type (vinyl-ester, epoxy, and polyester), and conditioning (with and without sustained load).

Declaration of competing interest

There are no conflicts of interest to disclose.

CRediT authorship contribution statement

Ahmed H. Ali: Data curation, Formal analysis, Validation, Writing - original draft. **Hamdy M. Mohamed:** Writing - review & editing. **Brahim Benmokrane:** Conceptualization, Supervision.

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References

- [1] Benmokrane B, Ali AH, Mohamed HM, Robert M, ElSafy A. Durability performance and service life of CFCC tendons exposed to elevated temperature and alkaline environment. *J Compos Construct* 2016. [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000606](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000606). 04015043.
- [2] Ali AH, Mohamed HM, Benmokrane B, ElSafy A, Chaallal O. Durability performance and long-term prediction models of sand-coated basalt FRP bars. *Compos Part B* 2019;157:248–58. 2019.
- [3] Wu G, Wang X, Wu ZS, Dong Q. Degradation of basalt FRP bars in alkaline environment. *Sci Eng Compos Mater* 2015;22(6):649–57.
- [4] Benmokrane B, Mohamed HM, Ali AH. Service-Life-Prediction and field application of GFRP tubular and solid bolts based on laboratory physical and mechanical assessment. *J Compos Mater* 2018;1–15. <https://doi.org/10.1177/0021998318764806>.
- [5] Hollaway LC. A review of the present and a 614 nd future utilisation of FRP composites in the civil infrastructure with reference to their important in-service properties. *Construct Build Mater* 2010;24:2419–45.
- [6] Wu Z, Wang X, Wu G. Advancement of structural safety and sustainability with basalt fiber reinforced polymers. In: *Proc., 6th int. conf. on FRP composites in civil engineering*. Rome, Italy: CICE; 2012. p. 29.
- [7] Wu G, Dong ZQ, Wang X, Zhu Y. Prediction of long-term performance and durability of BFRP bars under the combined effect of sustained load and corrosive solutions. *J Compos Construct* 2015;19(3). [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000517](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000517).
- [8] Wang ZK, Yang ZH, Yang YM, Xian GJ. Flexural fatigue behavior of a pultruded basalt fiber reinforced epoxy plate subjected to elevated temperatures exposure. *Polym Compos* 2016. <https://doi.org/10.1002/pc.24124>.
- [9] Mohamed HM, Ali AH, Benmokrane B. Mechanism of distributed composite GFRP bars in circular concrete members with and without spirals under shear. *J compos Part B* 2019;162:62–72.
- [10] Solyom S, Benedetti MD, Guadagnini M, Balázs GL. Effect of temperature on the bond behaviour of GFRP bars in concrete. *Composites Part B* 2020:183.
- [11] Ali AH, Mohamed HM, Benmokrane B, ElSafy A. Theory-based approaches and microstructural analysis to evaluate the service life-retention of stressed carbon fiber composite strands for concrete bridge applications. *Compos Part B* 2019;165: 279–92.
- [12] Ali AH, Mohamed, Benmokrane B, El-Safy A. Effect of applied sustained load and severe environments on durability performance of carbon-fiber composite cables (CFCCs). *J Compos Mater* 2018;1–16. <https://doi.org/10.1177/0021998318789742>.
- [13] Ali AH, Benmokrane B, Mohamed HM, Manalo A, El-Safy A. Statistical analysis and theoretical predictions of the tensile strength retention of GFRP bars based on resin degradation. *J Compos Mater* 2018;52(21):2929–48. <https://doi.org/10.1177/0021998318755866>.
- [14] Benmokrane B, Ali AH, Mohamed HM, El-Safy A, Manalo AC. Laboratory assessment and durability performance of vinyl-ester, polyester, and epoxy glass-FRP bars for concrete structures. *Composites Part B* 2017;114:163–74.
- [15] Robert M, Cousin P, Benmokrane B. Durability of GFRP reinforcing bars embedded in moist concrete. *J Compos Construct* 2009;66:66–73. [https://doi.org/10.1061/\(ASCE\)1090-0268\(2009\)13:2](https://doi.org/10.1061/(ASCE)1090-0268(2009)13:2).
- [16] Drouin, B., Latour, G., Mohamed, H. More than 10 years successful field applications of FRP bars in Canada. CDCC 2011, the fourth international conference on durability & sustainability of fiber reinforced polymer (FRP) composites for construction and rehabilitation, Quebec city, QC, Canada, 6 pp.

- [17] Liu T, Liu X, Feng P. A comprehensive review on mechanical properties of pultruded FRP composites subjected to long-term environmental effects. *Composites Part B* 2020;191:107958. <https://doi.org/10.1016/j.compositesb.2020.107958>.
- [18] Banibayat P, Patnaik A. Variability of mechanical properties of basalt fiber reinforced polymer bars manufactured by wet lay-up method. *Mater Des* 2014;56: 898–906.
- [19] Benmokrane B, Wang P, Ton-That T, Rahman H, Robert JF. Durability of glass fibre reinforcing bars in concrete environment. *J Compos Construct* 2002;6(3):143–53.
- [20] Montaigu M, Robert M, Ahmed E, Benmokrane B. Durability performance of new GFRP dowels for concrete pavement. *J Compos Construct* 2013;17(2):176–87.
- [21] ACI (American Concrete Institute). Specification for carbon and glass fiber-reinforced polymer bar materials for concrete reinforcement. ACI 2008;440. 6M-08, Farmington Hills, MI.
- [22] Canadian Standards Association (CSA). Specification for fibre reinforced polymers. CAN/CSA S807-10. 2010. Rexdale, Ontario, Canada.
- [23] Canadian Standards Association (CSA). Design and construction of building components with fibre-reinforced polymers. CAN/CSA S806-12. 2012. Rexdale, Ontario, Canada.
- [24] CSA (Canadian Standards Association). Canadian highway bridge design code." CAN/CSA-S6-14. 2014. Toronto.
- [25] Li C, Gao D, Wang Y, Tang J. Effect of high temperature on the bond performance between basalt fibre reinforced polymer (BFRP) bars and concrete. *Construct Build Mater* 2014;141:44–51.
- [26] Micelli F, Nanni A. Durability of FRP rods for concrete structures. *Construct Build Mater* 2004;18:491–503.
- [27] Elgabbas F, Ahmed E, Benmokrane B. Physical and mechanical characteristics of new basalt-FRP bars for reinforcing concrete structures. *Construct Build Mater* 2015;95:623–35.
- [28] Bank LC. *Composites for construction: structural design with FRP materials*. Hoboken: Wiley; 2006. <https://doi.org/10.1002/9780470121429>.
- [29] Holloway LC. Advanced fibre polymer composite structural systems used in bridgeengineering. *ICE Man Bridg Eng* 2008;503–29.
- [30] Benmokrane B, Elgabbas F, Ahmed EA, Cousin P. Characterization and comparative durability study of glass/vinylester, basalt/vinylester, and basalt/epoxy FRP bars. *J Compos Construct* 2015;19(6). [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000564](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000564).
- [31] Qin G, Na J, Mu W, Tan W, Yang J, Ren J. Effect of continuous high temperature exposure on the adhesive strength of epoxy adhesive, CFRP and adhesively bonded CFRP-aluminum alloy joints. *Composites Part B* 2018;154:43–55.
- [32] Serbescu A, Guadagnini M, Pilakoutas K. Mechanical characterization of basalt FRP rebars and long-term strength predictive model. *J Compos Construct* 2015;19(2). [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000497](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000497).
- [33] ACI (American Concrete Institute). Guide for the design and construction of structural concrete reinforced with FRP bars, vol. 440. Farmington Hills, MI: ACI; 2015. 1R-15.
- [34] ASTM. Standard test method for compositional analysis by thermogravimetry. West Conshohocken, PA: ASTM E1131; 2003.
- [35] ASTM Standard E1131-08. Standard test method for compositional analysis by thermogravimetry. ASTM E1131-08. West Conshohocken, Philadelphia, Pa: ASTM International; 2014. p. 19103.
- [36] ASTM Standard D570. Standard test method for water absorption of plastics. ASTM D570 – 98. West Conshohocken, Philadelphia, Pa: ASTM International; 2010. p. 19103.
- [37] ASTM Standard E1356. Standard test method for assignment of the glass transition temperatures by differential scanning calorimetry. ASTM E1356-08. West Conshohocken, Philadelphia, Pa: ASTM International; 2014. p. 19103.
- [38] ASTM. Tensile properties of fiber reinforced polymer matrix composite bars. West Conshohocken, PA: ASTM D7205; 2011.
- [39] ASTM. Standard test method for transverse shear strength of fiber-reinforced polymer matrix composite bars. West Conshohocken, PA: ASTM D7617; 2011.
- [40] ASTM. Standard test method for apparent horizontal shear strength of pultruded reinforced plastic rods by the short beam method. West Conshohocken, PA: ASTM D4475; 2008.
- [41] ASTM. Standard test method for flexural properties of fiber reinforced pultruded plastic rods. West Conshohocken, PA: ASTM D4476; 2009.
- [42] Patnaik A. Applications of basalt fiber reinforced polymer (BFRP) reinforcement for transportation infrastructure. *Dev Res Agenda Transport Infrastruct TRB* 2009; 5.
- [43] Benmokrane B, Manalo A, Bouhet JC, Mohamed K, Robert M. Effects of diameter on the durability of glass-1 fiber-reinforced-polymer (GFRP) bars conditioned in alkaline solution. *J Compos Construct* 2017;21(5). [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000814](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000814).
- [44] Yi S, Hilton HH. Effects of thermo-mechanical properties of composites on viscosity, temperature and degree of cure in thick thermosetting composite laminates during curing process. *J Compos Mater* 1998;32(7):600–22.
- [45] Cinquin J, Medda B. Influence of laminate thickness on composite durability for long term utilisation at intermediate temperature (100 - 150°C). *Compos Sci Technol* 2009;69:1432–6.
- [46] Kocaoz S, Samaranyake VA, Nanni A. Tensile characterization of glass FRP bars. *Composites Part B* 2005;36:127–34.
- [47] Portnov G, Bakis CE. Analysis of stress concentration during tension of round pultruded composite rods. *Compos Struct* 2008;83:100–9.
- [48] Wisnom MR, Jones MI. Size effects in interlaminar tensile and shear strength of unidirectional glass fibre/epoxy. *J Reinforc Plast Compos* 1996;15(1):2–15.
- [49] Koller R, Chang S, Xi Y. Fiber-reinforced bars under freeze-thaw cycles and different loading rates. *J Compos Mater* 2007;41(1):5–25.
- [50] Carvelli V, Giulia F, Pisani MA. Anchor system for tension testing of large diameter GFRP bars. *J Compos Construct* 2009;13(5):344–9.
- [51] Nkurunziza G, Debaiky A, Cousin P, Benmokrane B. Durability of GFRP bars: a critical review of the literature. *Prog Struct Eng Mater* 2005;7(4):194–209.
- [52] Archillides Z, Pilakoutas K. Bond behaviour of fibre reinforced polymer bars under direct pullout conditions. *J Compos Construct* 2004;8(2):173–81.
- [53] Castro PF, Carino NJ. Tensile and non-destructive testing of FRP bars. *J Compos Construct* 1998;2(1):17–27.
- [54] Maranan GB, Manalo AC, Karunasena W, Benmokrane B, Lutze D. Flexural behaviour of glass fibre reinforced polymer (GFRP) bars subjected to elevated temperature. In: Smith ST, editor. 23rd australasian conference on the mechanics of structures and materials (ACMSM23) byron bay, Australia; December 2014. p. 9–12.